

Current-Voltage Characteristics

□ Qualitative Analysis: Thermal Equilibrium

Electrons

Diffusion

- Electron diffusion from n-side to p-side
- Most will be reflected
- Few electrons will have enough energy and overcome the barrier height

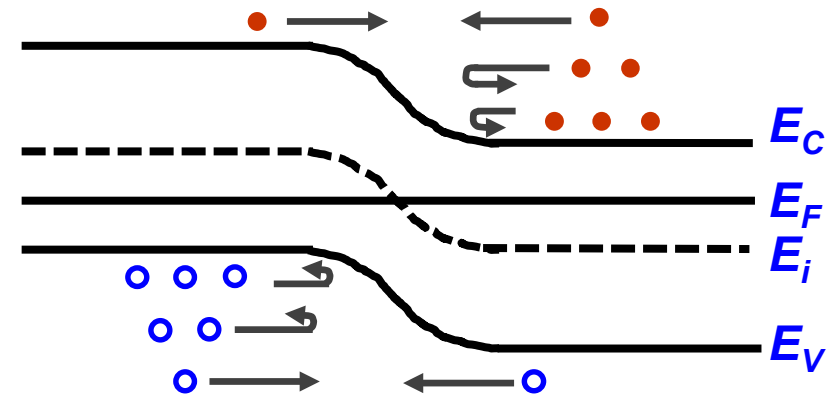
Drift

- Electron drift from p-side to n-side
- Few electrons will be drifted (e- is minority carriers in p-side)
- At thermal equilibrium, diffusion and drift are balanced

Net electron current = 0

Similarly, net hole current=0

Thus, the total current=0



Equilibrium ($V_A = 0$)

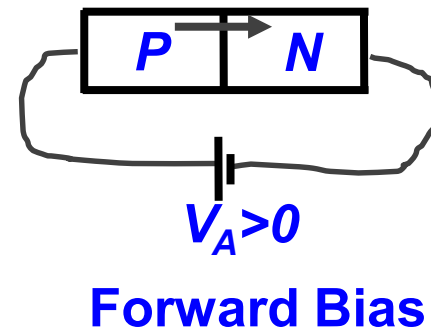
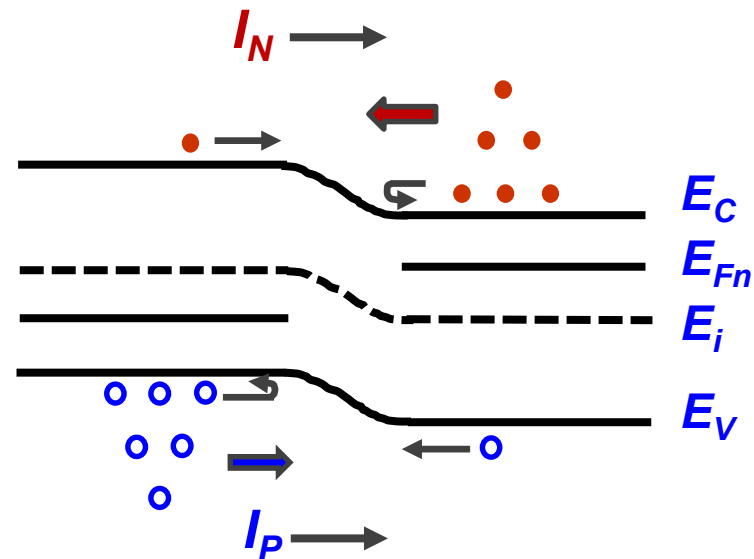
Particle Flow		Current
←	Electron Diffusion	→
→	Electron Drift	←
→	Hole Diffusion	→
←	Hole Drift	←

Current-Voltage Characteristics

□ Qualitative Analysis: Forward Bias ($V_A > 0$)

Electrons

- More electron diffusion from n-side to p-side due to lowering of barrier height
- Electrons drift from p-side to n-side remains same (minority concentration unchanged)
- Electron diffusion dominates
- Diffusion current increases exponentially with V_A
 - ✓ Potential barrier ($V_{bi} - V_A$) linearly decrease with increasing V_A
 - ✓ Plus electron concentration (in n-side, majority carrier) above the barrier height increases exponentially when the energy level moves upward



Current-Voltage Characteristics

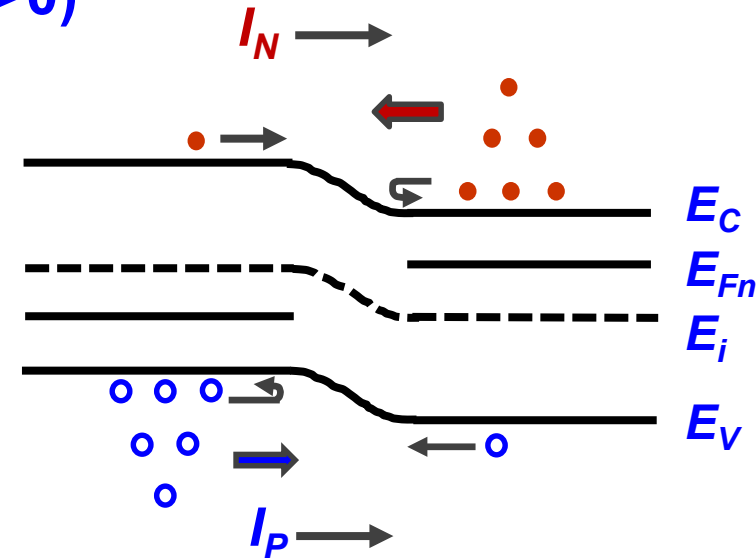
□ Qualitative Analysis: Forward Bias ($V_A > 0$)

- Similar analysis applied to holes
- The total current is dominated by both electron and hole diffusion and given as

$$J_F = J_{N, F} + J_{P, F}$$

- Forward current increases exponentially with V_A

Will V_A be larger than V_{bi} ?



Particle Flow		Current
←	Electron Diffusion	→
→	Electron Drift	←
→	Hole Diffusion	→
←	Hole Drift	←

Current-Voltage Characteristics

□ Qualitative Analysis: Reverse Bias ($V_A < 0$)

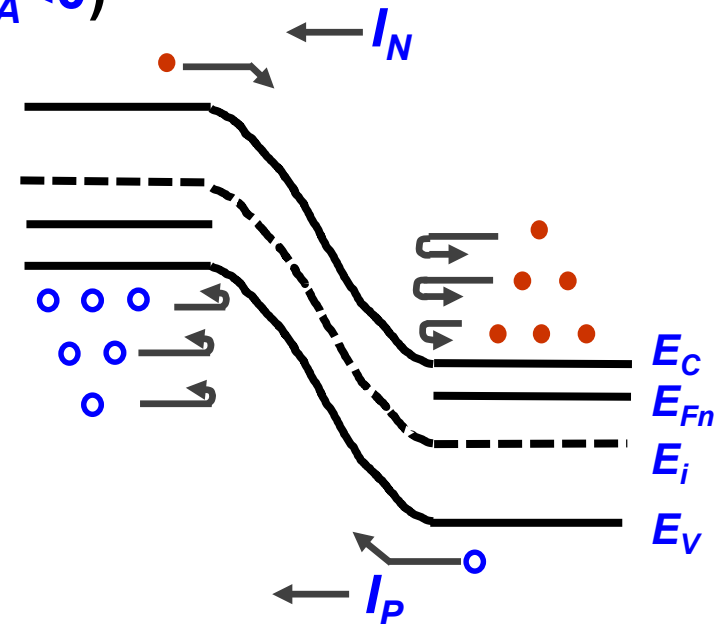
Electrons

- Very few electron diffusion from n-side to p-side due to increased barrier height
- Electrons drift from p-side to n-side remains same (minority concentration unchanged)
- Electron current is dominated by drift
- Thus electron current is very small

Similar analysis applied to holes

So, the total reverse current dominated by both electron and hole drift currents

Reverse currents are very small and nearly independent with V_A value

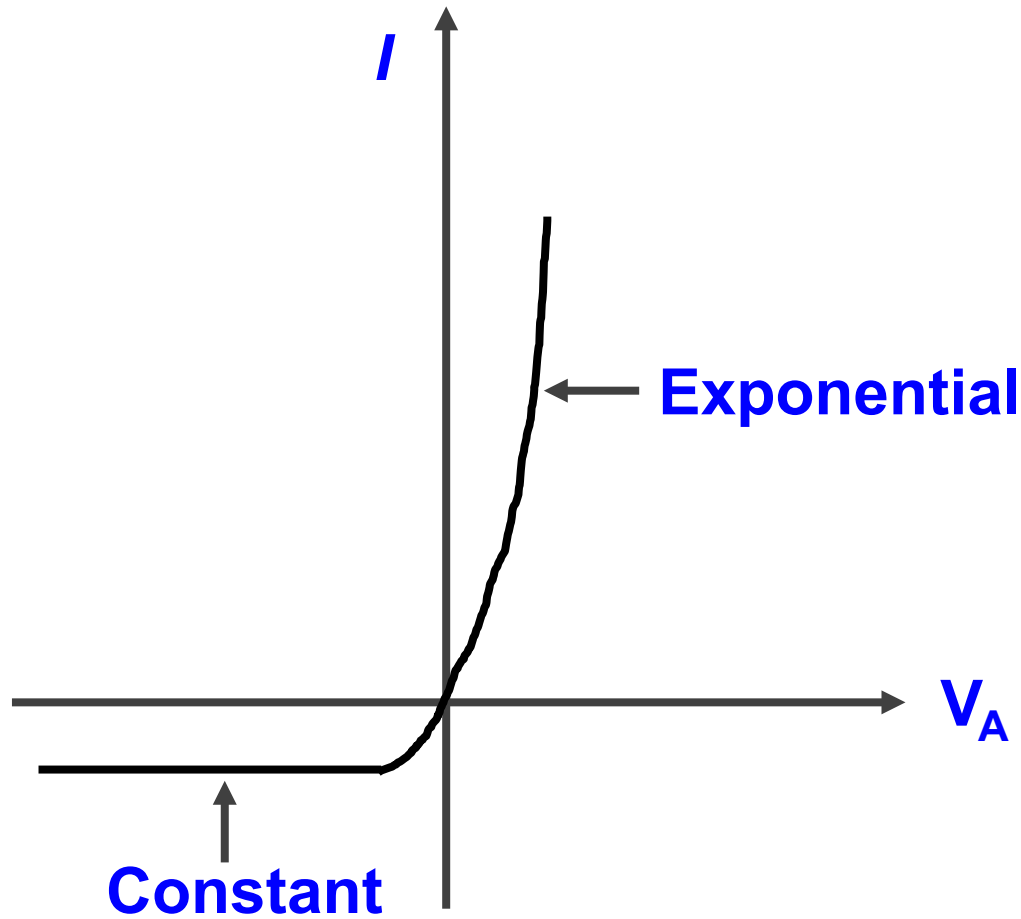


Particle Flow		Current
←	Electron Diffusion	→
→	Electron Drift	←
→	Hole Diffusion	→
←	Hole Drift	←

Reverse Bias

Current-Voltage Characteristics

□ Qualitative Analysis: I - V Characteristics



Current-Voltage Characteristics

□ Quantitative Analysis: I - V Characteristics

Assumptions

- (1) The diode is being operated under steady state conditions
- (2) The doping profile: a non-degenerately doped step junction models
- (3) The diode is one dimensional
- (4) Low level injection prevails in the quasi-neutral regions
- (5) There are no processes other than drift, diffusion, and thermal R-G (with negligible) taking place inside the diode.

Thus, the current can be calculated as

$$I = AJ$$

$$J = J_N(x) + J_P(x)$$

$$J_N(x) = q\mu_n nE + qD_N \frac{dn}{dx}$$

$$J_P(x) = q\mu_p pE - qD_p \frac{dp}{dx}$$

Current-Voltage Characteristics

Quantitative Analysis: I - V Characteristics

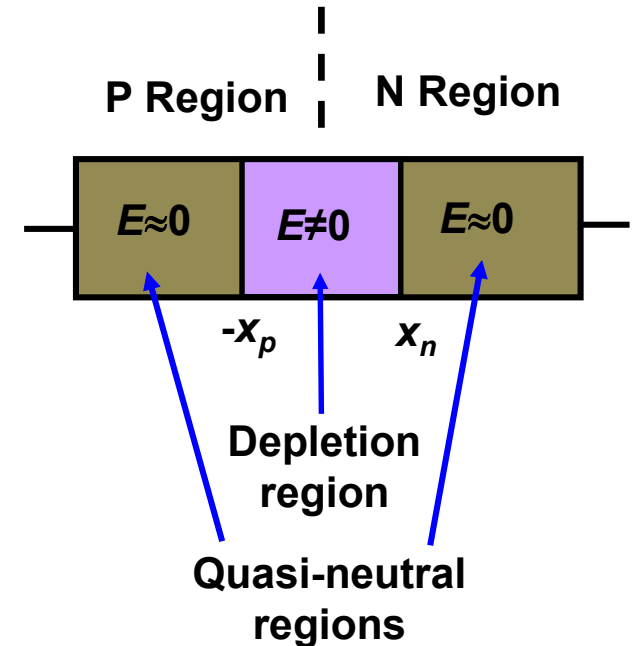
Depletion region

- Recall $0 = \frac{1}{q} \frac{dJ_N}{dx} + \frac{\partial n}{\partial t} \Big|_{R-G}$

$0 = -\frac{1}{q} \frac{dJ_P}{dx} + \frac{\partial p}{\partial t} \Big|_{R-G}$

- From assumption (5), we have $\left\{ \begin{array}{l} \frac{\partial n}{\partial t} \Big|_{R-G} = 0 \\ \frac{\partial p}{\partial t} \Big|_{R-G} = 0 \end{array} \right.$

- Then, we have $\left\{ \begin{array}{l} \frac{dJ_N}{dx} = 0 \\ \frac{dJ_P}{dx} = 0 \end{array} \right.$



- Which means that J_N and J_P are constant within the depletion region

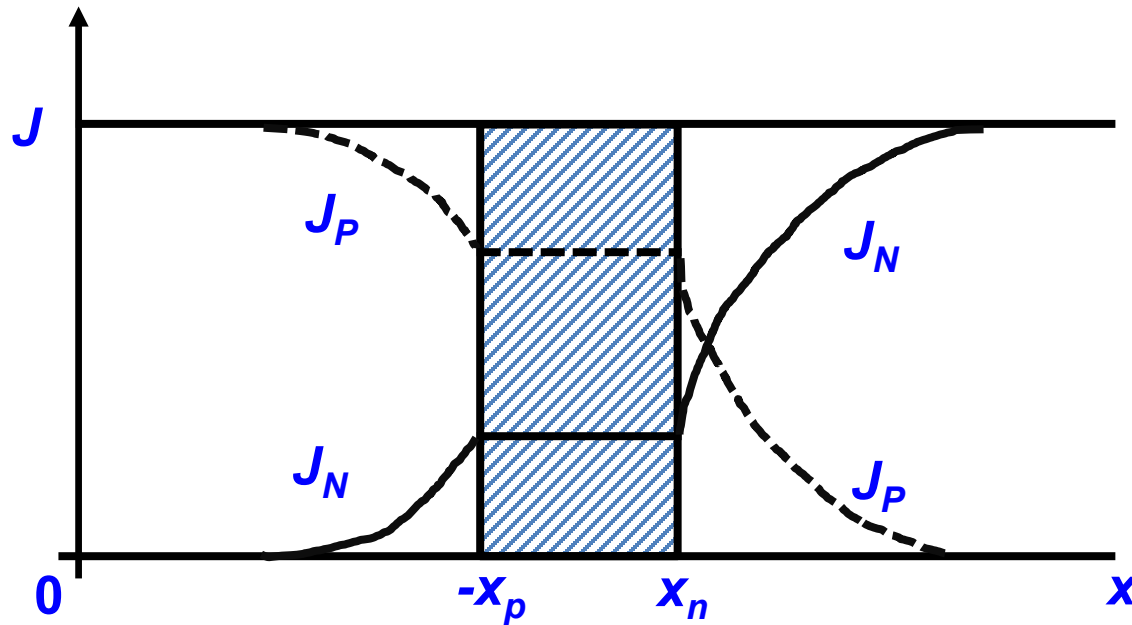
$$J_N(-x_p \leq x \leq x_n) = J_N(-x_p)$$

$$J_P(-x_p \leq x \leq x_n) = J_P(x_n)$$

Current-Voltage Characteristics

□ Some Useful Discussions

Carrier Current Components



Think: How do we calculate the current?

Current-Voltage Characteristics

Quantitative Analysis: I - V Characteristics

Depletion region

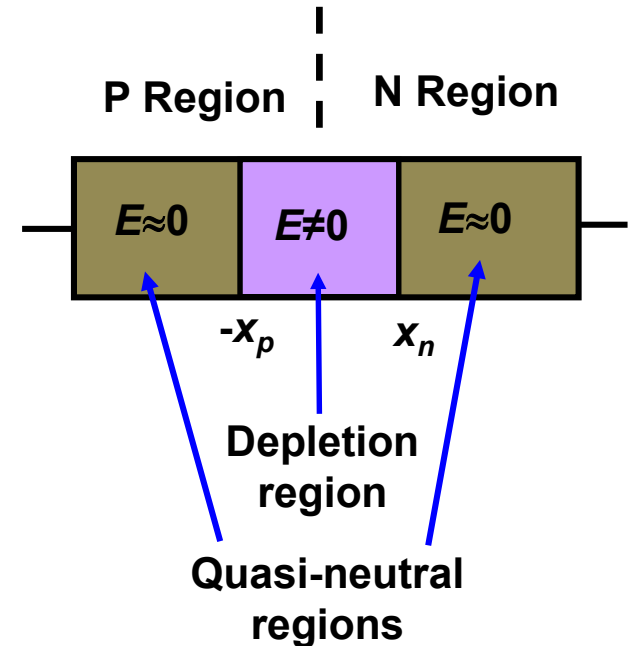
- Since J_N and J_P are constant within depletion region

$$J_N(-x_p \leq x \leq x_n) = J_N(-x_p)$$

$$J_P(-x_p \leq x \leq x_n) = J_P(x_n)$$

- The total current density can be given as below

$$J = J_N(x) + J_P(x) = J_N(-x_p) + J_P(x_n)$$



Thus, the total current density can be calculated as the sum of diffusion currents at both depletion edges $J_N(-x_p)$ and $J_P(x_n)$!!!

Current-Voltage Characteristics

Quantitative Analysis: I - V Characteristics

Quasi-neutral regions

- Since low level injection ($E \approx 0$) and $\frac{dn_0}{dx} = \frac{dp_0}{dx} = 0$

Thus, current density in these two quasi-neutral region can be calculated as

$$J_N(x) = qD_N \frac{d\Delta n_p}{dx} \quad \text{For } x \leq -x_p$$

$$J_P(x) = -qD_P \frac{d\Delta p_n}{dx} \quad \text{For } x \geq x_n$$

Since low-level injection, the majority carrier concentration remains the same within quasi-neutral regions, thus the E -field is almost zero.

Recall:

$$J_p = J_{p,drift} + J_{p,diff} = q\mu_p pE - qD_p \nabla p$$

$$J_n = J_{n,drift} + J_{n,diff} = q\mu_n nE + qD_n \nabla n$$

Thus, we are able to calculate the current density (diffusion current) at quasi-neutral regions!!!

Current-Voltage Characteristics

□ Quantitative Analysis: *I*-*V* Characteristics

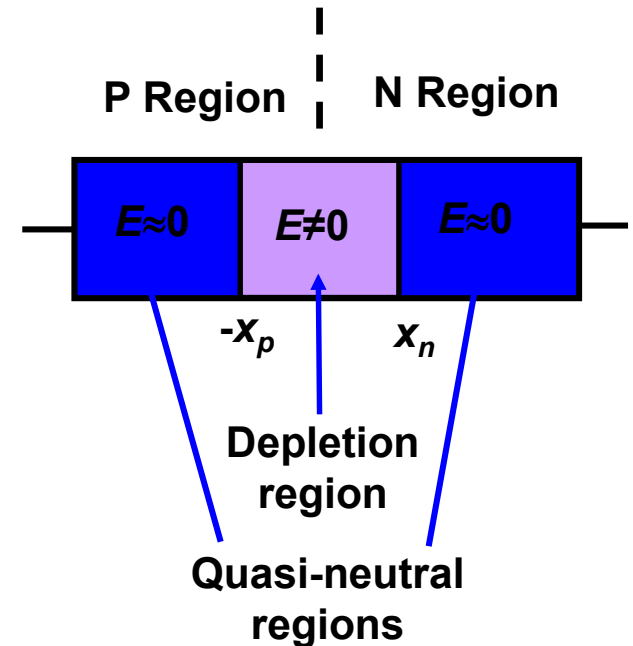
Quasi-neutral regions

- The minority carrier diffusion equations are

$$0 = D_N \frac{d^2 \Delta n_p}{dx^2} - \frac{\Delta n_p}{\tau_n} \quad x \leq -x_p$$

$$0 = D_P \frac{d^2 \Delta p_n}{dx^2} - \frac{\Delta p_n}{\tau_p} \quad x \geq x_n$$

From the above two equations (plus boundary condition, will be discussed later), both Δn_p and Δp_n can be obtained



Current-Voltage Characteristics

Quantitative Analysis: I - V Characteristics

Boundary Conditions

At the Ohmic contacts:

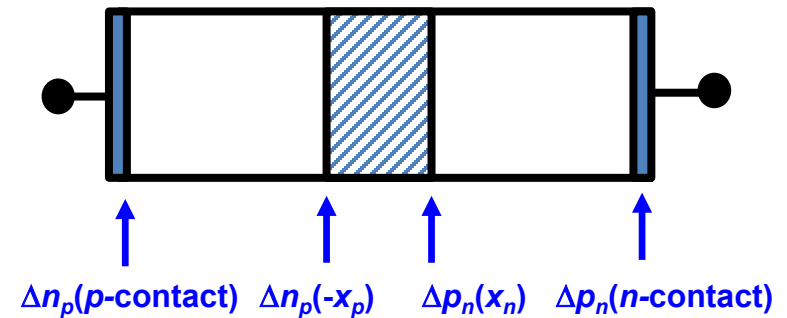
- Consider long “wide” base diode, which means the diode has contacts which are larger than the minority carrier diffusion lengths. The contacts can be viewed at

$$x = \pm\infty$$

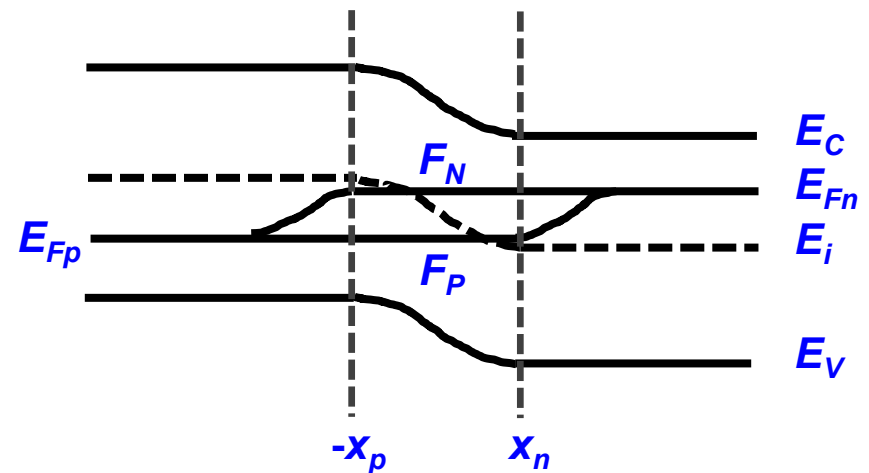
- So, the boundary conditions are

$$\Delta n_p(x = -\infty) = 0$$

$$\Delta p_n(x = \infty) = 0$$



(a)



(b)

Current-Voltage Characteristics

Recall:

$$n = n_i e^{(F_N - E_i)/kT}$$

$$p = n_i e^{(E_i - F_p)/kT}$$

Quantitative Analysis: I - V Characteristics

Boundary Conditions

At Depletion Region Edges

- Introducing “quasi-Fermi level” F_N and F_P ($F_N - F_P = V_A$). Inside depletion region, we have

$$np = n_i^2 e^{qV_A/kT}$$

- So, the boundary conditions are

$$p_p(-x_p) = N_A \quad n_p(-x_p) = \frac{n_i^2}{N_A} e^{qV_A/kT}$$

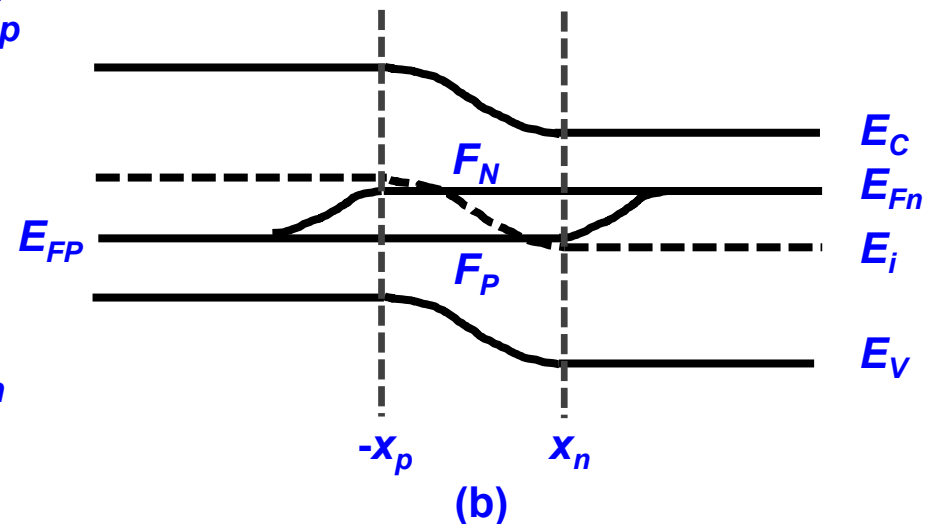
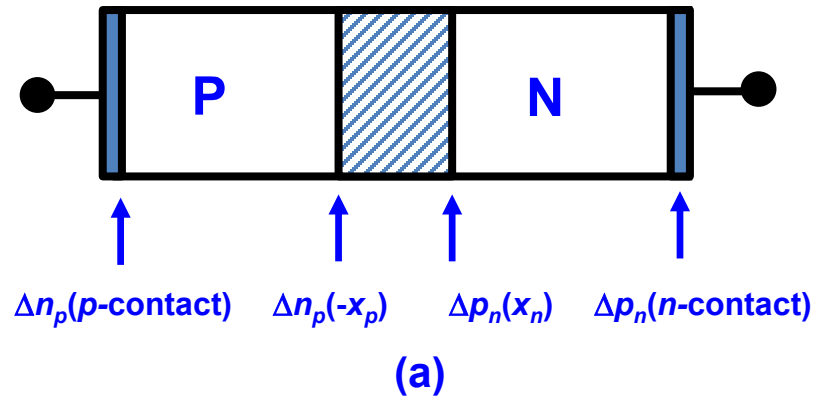
$$\Delta n_p(-x_p) = \frac{n_i^2}{N_A} (e^{qV_A/kT} - 1)$$

at $x = -x_p$

$$n_n(x_n) = N_D \quad p_n(x_n) = \frac{n_i^2}{N_D} e^{qV_A/kT}$$

$$\Delta p_n(x_n) = \frac{n_i^2}{N_D} (e^{qV_A/kT} - 1)$$

at $x = x_n$

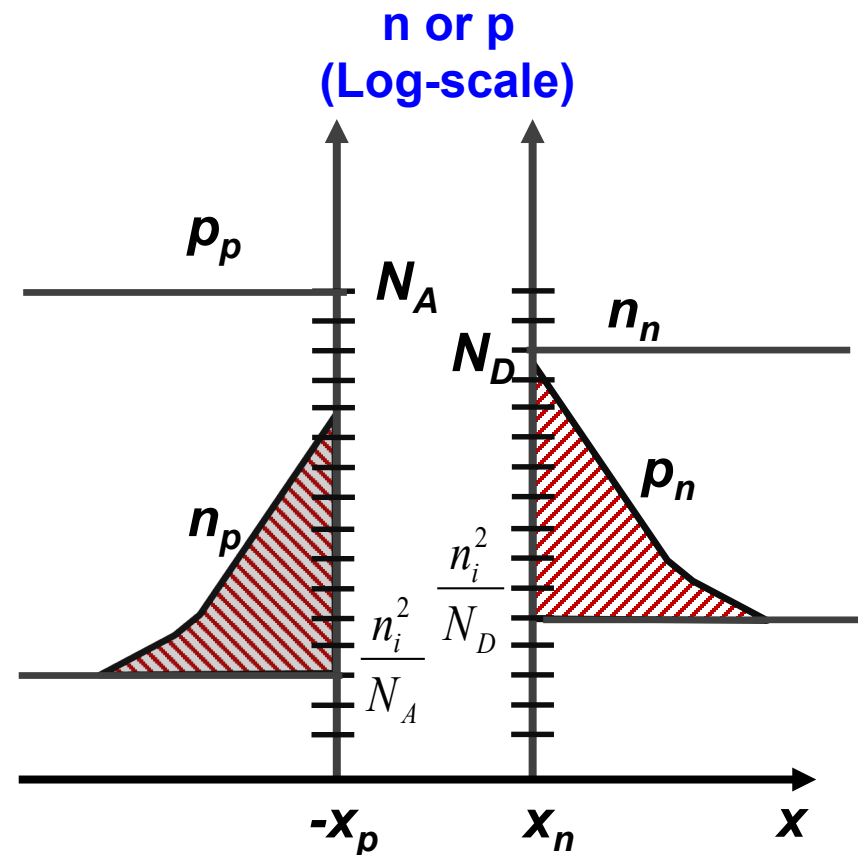


Current-Voltage Characteristics

□ Some Useful Discussions

Carrier Concentrations (Forward bias)

- Carriers are injected into other side by diffusion and become minority carriers.
- The excess minority carriers are eliminated by recombination during diffusion deeper into the regions
- Thus, there is a build-up of minority carrier in the quasi-neutral regions immediately adjacent to the edges of the depletion regions.



Current-Voltage Characteristics

Quantitative Analysis: I - V Characteristics

Calculation of Current $J_P(x_n)$

- (1) Shift the origin of coordinates to x_n
- (2) The equation and boundary conditions are

$$0 = D_p \frac{d^2 \Delta p_n}{dx'^2} - \frac{\Delta p_n}{\tau_p} \quad \mathbf{x' \geq 0 \quad \text{Continuity Equation}}$$

Subject to the boundary conditions

$$\Delta p_n(x' = \infty) = 0 \quad \mathbf{\text{Ohmic Contact}}$$

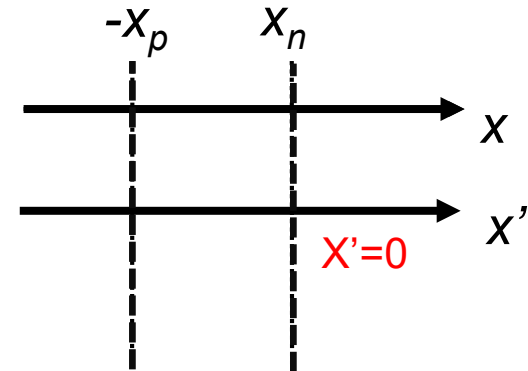
$$\Delta p_n(x' = 0) = \frac{n_i^2}{N_D} (e^{qV_A/kT} - 1) \quad \mathbf{\text{Law of Injection}}$$

- (3) Solving the above equations, we have

$$J_P(x') = -qD_p \frac{d\Delta p_n}{dx'} = q \frac{D_p n_i^2}{L_p N_D} (e^{qV_A/kT} - 1) e^{-x'/L_p}$$

- (4) So, $J_P(x_n)$ can be given as

$$J_P(x = x_n) = J_P(x' = 0) = q \frac{D_p n_i^2}{L_p N_D} (e^{qV_A/kT} - 1)$$



Current-Voltage Characteristics

□ Quantitative Analysis: I - V Characteristics

Calculation of Current

(1) Similarly, $J_N(-x_p)$ can be obtained as follow:

$$J_N(x = -x_p) = q \frac{D_n n_i^2}{L_n N_A} (e^{qV_A/kT} - 1)$$

(2) So, the total current is

$$I = AJ = qA \left(\frac{D_p n_i^2}{L_p N_D} + \frac{D_n n_i^2}{L_n N_A} \right) (e^{qV_A/kT} - 1)$$

(3) A useful and important equation:

$$I = I_0 (e^{qV_A/kT} - 1)$$

$$I_0 \equiv qA \left(\frac{D_p n_i^2}{L_p N_D} + \frac{D_n n_i^2}{L_n N_A} \right)$$

Current-Voltage Characteristics

□ Some Useful Discussions

I-V Characteristics (Ideal)

- (1) For reverse biases greater than a few kT/q ,

$$I = -I_0$$

- (2) For forward biasing larger than a few kT/q ,

$$I \rightarrow I_0 \exp(qV_A / kT)$$

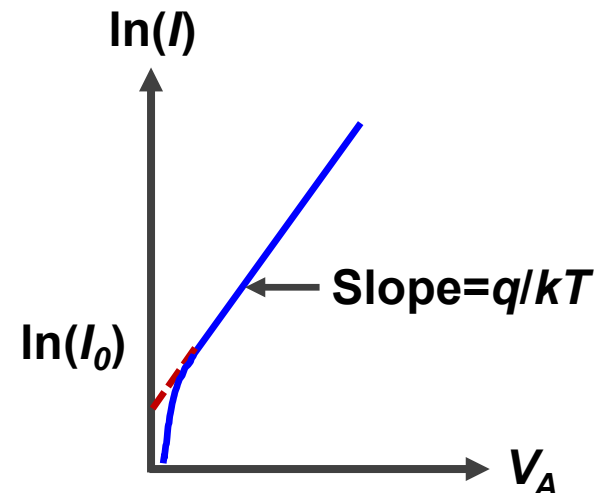
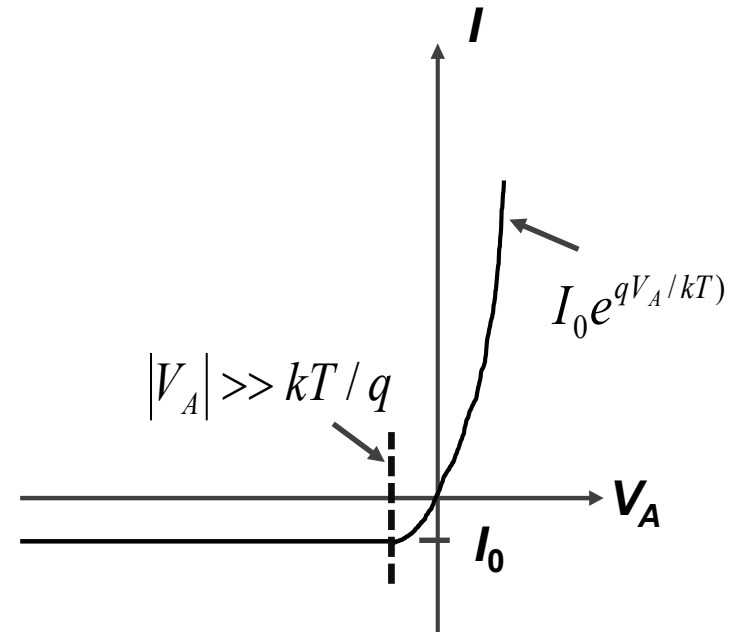
- (3) In forward bias conditions, I - V is always plotted as semi-log scale since

$$\ln(I) = \ln(I_0) + \frac{q}{kT} V_A$$

- (4) Saturation current

$$I_0 \equiv qA \left(\frac{D_p n_i^2}{L_p N_D} + \frac{D_n n_i^2}{L_n N_A} \right)$$

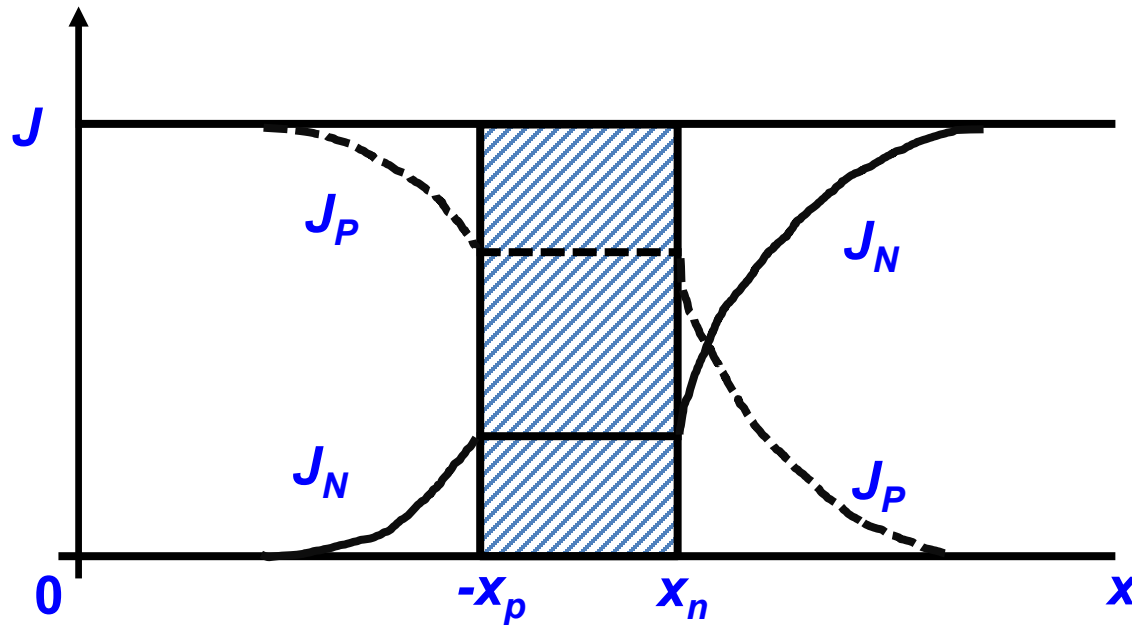
- depends on intrinsic concentration
- depends on doping concentration



Current-Voltage Characteristics

□ Some Useful Discussions

Carrier Current Components



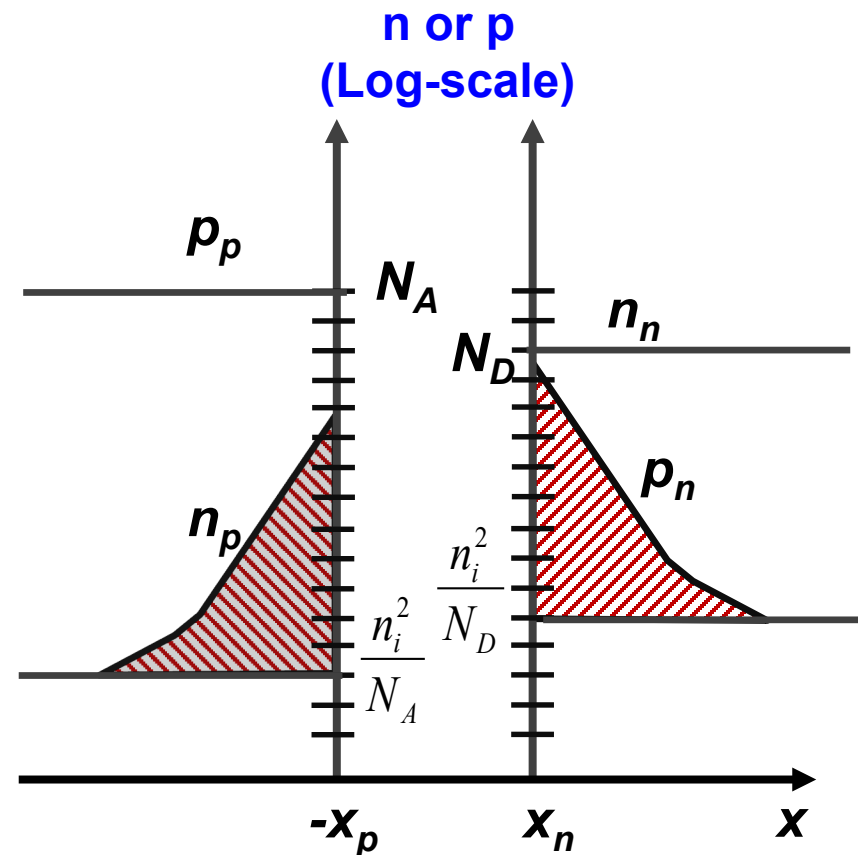
Think Again: How do we calculate the current?

Current-Voltage Characteristics

□ Some Useful Discussions

Carrier Concentrations (Forward bias)

- Carriers are injected into other side by diffusion and become minority carriers.
- The excess minority carriers are eliminated by recombination during diffusion deeper into the regions
- Thus, there is a build-up of minority carrier in the quasi-neutral regions immediately adjacent to the edges of the depletion regions.



Current-Voltage Characteristics

□ Some Useful Discussions

Carrier Concentrations (Reverse bias)

- The depletion region acts like a “sink” for minority carriers, draining the carriers from the adjacent quas-neutral region.
- A reverse bias of a few kT/q effectively reduces the minority carrier concentrations to zero at the edges of the depletion region.
- Larger reverse biases have little effect on the carrier concentration (consistent with the saturation current in reverse bias).

